

PROBLEMAS 1

* CASO 1

$lw, (rs, LSR\ inm_{0..1})$

accede a la direccion pos rs , actualiza Rs para la siguiente acción

$lw\ rt, [rs] \#inm:$

$BR(rt) \leftarrow Mem(BR(rs)):$

$BR(rs) \leftarrow BR(rs) + SignExt(inm_{16});$

$PC \leftarrow PC + 4$

① Fetch

$IR \leftarrow M(PC)$

$PC \leftarrow PC + 4$

$PcWrite = 1$

$ALUsrcA = 1$

$ALUout = 1$

$Mwrite = 0$

$ALUsrcB = 3$

② Decode

$A \leftarrow BR(rs)$

$B \leftarrow BR(rt)$

$RagWrite = 0$

$Awrite = 1$

$Bwrite = 1$

$PcWrite = 0$

③ Execution

$ALUout \leftarrow A + sign(inm)$

$ALUsrcA = 0$

$Pcwrite = 0$

④ Memory

$MDR \leftarrow M(ALUout)$

$BR(rs) \leftarrow ALUout$

$ALUop = 1$

$Pcwrite = 0$

⑤ Write

$BR(rt) \leftarrow MDR$

$OutWrite = 1$

* C650 3

ret rs

$PC \leftarrow BR(rs)$

(1)

$IR \leftarrow M(PC)$

PCWrite = 0
IRWrite = 1
ALU out = 1
MWrite = 0

(2)

$A \leftarrow BR(rs)$

$B \leftarrow BR(rl)$

RegWrite = 0
AWrite = 1
BWrite = 1
PCWrite = 0

(3)

$PC \leftarrow M(A)$

PCWrite = 1
OutWrite = 1

